1. Can you describe your role in relation to the student research projects you have been involved in or supervised?

I lead and coordinate the Value Chain Hackers initiative and I am closely involved in how student research projects are set up, executed, and handed over. My role sits between infrastructure, research practice, and supervision. I am often the one trying to ensure continuity across projects and to make outputs usable beyond the individual student group. Because of that position, I also see very clearly where things break down.

2. At the start of projects, what guidance are students typically given about how they are expected to document their research work? Can you give an example?

The guidance is generally too vague. Students are told that documentation is important, that they should store their work somewhere centrally, and that there needs to be a final report. However, there is rarely concrete instruction on *what* needs to be documented, *how*, or *why*. However, in the beginning we do tell them to utilise "GitHub" or "use a shared drive," but without defining expectations around decision logs, raw data, analysis steps, or reproducibility. As a result, students interpret documentation as "write a report at the end." with no history, no logging, so I as an assessor cannot discern whether they made this in one evening, last minute, or that this has been a process of many months with many individuals. This is specifically dangerous when considering the new trend of A.I usage for the students which I know they use, because we specifically give them access to the tools. As such the only way I can collaborate history or see how they got their information is by going through the A.I tools and then seeing what they actually asked the a.i to do for them.

Here is the A.I stack that we gave the students access to:
https://github.com/Windesheim-A-I-Support/InstallLocalAiPackage

3. Can you describe specific situations where documentation came up during project work with students? What happened in those moments?

Documentation usually becomes an issue when something goes wrong. For example, when a stakeholder asks how a conclusion was reached, or when a new supervisor or student joins the project. In those moments, it becomes clear that decisions were made verbally, analysis happened implicitly, and key context exists only in people's heads. We then try to reconstruct reasoning after the fact, which is inefficient and unreliable.

4. Based on the final outputs of projects, how usable do you think they are for someone new who has not spoken to the original student team? Can you give an example?

In most cases, they are **not very usable**. They are readable, but not reproducible. A new person can understand *what* the outcome was, but not *how* it was produced. For example, a report might state conclusions based on interviews, but the interview notes, selection logic, and interpretation steps are missing or inaccessible. Without talking to the original students, it is often impossible to confidently reuse the work.

5. Which parts tend to be difficult for someone new to understand based only on what is written down or stored?

The most difficult parts are:

- The context in which decisions were made
- Why certain paths were chosen and others dropped
- How raw inputs were transformed into conclusions

These elements are essential for trust, but they are exactly the parts that are usually undocumented.

6. Can you think of an example of an important decision or change during a project, and describe how and where it was documented, if at all?

Scope changes are a common example. Projects frequently change direction after early insights or stakeholder feedback. These decisions are discussed in meetings and sometimes agreed verbally, but rarely documented explicitly. Instead, the change only becomes visible indirectly in later versions of the report. There is almost never a clear record of *when* the decision was made, *by whom*, and *for what reason*.

7. If someone new wanted to understand how one important conclusion of a project was reached, what would they need to look at or read?

In theory, they would need access to raw data, analysis notes, intermediate drafts, and the final report. In practice, those elements are scattered across tools and personal storage, or missing altogether. There is no single traceable chain from evidence to conclusion, which makes verification and reuse very difficult.

8. During projects, is it clear who is responsible for documenting what, if anyone? How is that responsibility typically communicated or decided?

No, this is usually not clear. Responsibility for documentation is implicit and diffuse. Everyone is assumed to document their own work, but no one is explicitly responsible for documenting decisions, maintaining structure, or ensuring consistency. Because of that, documentation becomes incidental rather than intentional.

9. What tools, templates, or systems are students expected to use to document work, and where is everything actually stored in practice?

There is no consistent standard. Students are expected to use common tools like Word, Google Docs, or sometimes GitHub, but the actual practice is fragmented. Files are spread across shared drives, personal laptops, messaging apps, and visual tools like Miro. There is rarely a clear “source of truth,” and version control is often weak or absent.

10. What do you think the purpose of documentation is in these projects? Who do you think benefits from it, if anyone? And what, if anything, influences how motivated students are to document their work well?

The purpose of documentation should be reproducibility, trust, and continuity. In applied research, outputs should be inspectable and reusable. In reality, documentation is often treated as an administrative requirement rather than a core research activity. Students are less motivated because documentation is not clearly integrated into the workflow and because they are not consistently rewarded or assessed on the quality of their research trail.

11. Based on your experience, what tends to get lost most often over time across projects, for example context, decisions, data, analysis steps, or contacts?

What gets lost most often is:

- Decision rationale
- Analysis steps
- Context about constraints and changes

These losses make it very difficult to build on previous work without repeating mistakes or redoing research.

12. How does documentation generally fit into project work from your perspective? Does it seem integrated into the work, or more like something added on top? Why?

It is largely added on top. The primary focus is on delivering a final product, not on maintaining a transparent research process. Because supervision and assessment also focus mainly on the final output, there is little structural incentive to document the process properly.

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13. From your perspective, what currently makes it difficult to achieve consistent documentation across projects, and what would need to be in place for research outputs to be considered trustworthy and usable over time?

The main problems are unclear standards, fragmented tooling, and lack of ownership. To improve this, we would need explicit documentation requirements, shared infrastructure, and a clear definition of what “reproducible enough” means in applied student research. Trustworthy outputs require that a third party can trace a conclusion back to its underlying evidence without relying on oral explanation.

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14. Is there anything about documentation, continuity, or research practice that we have not discussed but that you think is important?

Yes. Without reproducibility, we are not really doing research — we are producing narratives. If Value Chain Hackers wants to claim research value, it needs to take documentation and reproducibility seriously as first-class concerns. Otherwise, knowledge remains locked in people instead of becoming a shared asset.